```
def Accept_data(A):
        n=int(input("Enter the number of Account ID's:"))
        for i in range(n):
        ID=int(input(f"Enter Account ID's: "))
        A.append(ID)

def Display_data(A):
        print("Customer ID's are:")
        a=len(A)
        for i in range(a):
        print(A[i])

def linear_search(arr,target):
        for index in range(len(arr)):
        if arr[index]==target:
        return index
        return -1

def binary_search(arr,target):
        low=0
        high=len(arr)
        while low<=high:
        mid=(low+high)//2
        if arr[mid]==target:
        return mid
        elif arr[mid]<target:
        low=mid+1
        elif arr[mid]>target:
        high=mid-1
        return -1


def main():
        Cust_ID=[]
        while True:
        print("\n 1.Customer ID's")
        print("\n 2.Linear search")
        print("\n 3.Binary search")
        print("\n 4.End")
        ch=int(input("Enter your choice:"))
        if ch==1:
        print("Enter the Customer ID's")
        Accept_data(Cust_ID)
        Display_data(Cust_ID)
```

```
        elif ch==2:
        print("Performing Linear Search")
        target=int(input("Enter the number to be search: "))
        result=linear_search(Cust_ID,target)
        if result!=-1:
                print(f"Element found at index {result}")
        else:
                print(f"Element not found")
        elif ch==3:
        print("Performing Binary Search")
        target=int(input("Enter the number to be search: "))
        result=binary_search(sorted(Cust_ID),target)
        if result!=-1:
                print(f"Element found at index {result}")
        else:
                print(f"Element not found")
        target
        elif ch==4:
        print("End")
        quit()
        else:
        print("Invalid choice!")
main()
```

Output:

```
1.Customer ID's
2.Linear search
3.Binary search
4.End
Enter your choice:
1
Enter the Customer ID's
1001
1003
1002
1006
1004
Enter your choice:
2
Performing Linear Search
Enter the number to be searh:
1002
Element found at index 2
Enter your choice:
1.Customer ID's
2.Linear search
3.Binary search
4.End
Enter your choice:
```